MILITARY SPECIFICATION

LAUNCHER, ROCKET LAU-10/A

This specification has been approved by
the Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers a dual purpose airborne external store type rocket launcher capable of shipping, storing and launching four 5.0-inch Folding Fin Aircraft Rockets. This launcher has two configuration as follows:

Configuration "A" - Shipping Configuration

Configuration "B" - Flight Configuration

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATION

Military

MIL-D-5028	Drawings and Data Lists: Preparation of Manufacturer's (for Production Aeronautical and Associated Equipment)
MIL-C-5015	Connectors, Electrical, "AN" type
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment; General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical Preparation for Delivery
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft
MIL-A-8591	Airborne Stores and Associated Suspension Equipment; General Design Criteria for
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment

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STANDARDS

Military

MIL-STD-105 and appendix	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage •

DRAWINGS

Bureau of Aeronautics

55A27J1	Launcher, Rocket LAU-10/A (including data list, all drawings and specifications listed thereon)

PUBLICATIONS

Air Force - Navy Aeronautical Bulletins

No. 143	Specifications and Standards; Use of

2.2 Availability of Documents:

(1) Give both the title and number when requesting applicable documents. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable documents may be obtain upon application to the Commanding Officer, Naval Aviation Supply Depot, (CDS), Philadelphia 11, Pennsylvania.

2.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, precedence shall be as follows:

(1) Contract - The contract shall have precedence over all referenced documents.

(2) This Specification - This specification shall have precedence over all applicable subsidiary documents. Any deviation from this specification, or from applicable subsidiary documents shall be specifically approved in writing by the procuring agency.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials -

3.1.1 Materials, Parts and Processes - Materials, parts and processes shall conform to applicable specifications as specified herein. Materials which are not covered by applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended. Specifications and standards for all materials, parts and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with Bulletin No. 143.

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3.1.2 Manufacturing Processes and Finish - Unless otherwise approved in writing by the procuring service, the manufacturing processes and finishes shall be as specified herein or on the applicable drawing or drawings.

3.2 Design and Construction:

3.2.1 Conformance to Drawings and Specifications - The launcher shall be constructed in accordance with the applicable drawings and specifications listed in Drawing 55A27J1. No departure from, revision or change to the drawings or specifications listed therein shall be made without prior written approval by the procuring service.

3.2.2 Interchangeability - All parts having the same part number shall be directly and completely interchangeable with each other in respect to fit, function and performance. All applicable provisions of Military Specification MIL-I-8500 shall apply.

3.2.3 Nomenclature and Nameplates - Nomenclature and nameplates shall be in accordance with Specification MIL-N-18307.

3.2.4 Weight - The weight of the launcher in Configuration B including all attachments and less rockets shall not exceed 106 pounds.

3.2.5 Electrical Connectors - All electrical connectors shall conform to Specification MIL-C-5015.

3.3 Performance - In addition to the above requirements, the equipment shall meet all requirements imposed by the tests of Section 4.

3.4 Workmanship - All details of workmanship shall be in accordance with high quality armament manufacturing practice. All parts shall be free from excessive marks, grooves, indentations, cracks, flaws and any other defects which may impair the serviceability, function or performance of the assembled unit. All workmanship shall be of a quality satisfactory to the procuring service or direct representative thereof.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine compliance with all the applicable requirements.

(1) Preproduction Tests

(2) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be conducted on ten equipments representative of the equipments to be supplied under the contract. Preproduction tests shall be conducted at a laboratory designated by the Bureau of Aeronautics. Preproduction tests shall consist of all tests specified in 4.4 and any other tests deemed necessary by the Bureau of Aeronautics to determine conformance with this specification.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing individual acceptance tests. The launchers shall be submitted for acceptance by lots. Equipments comprising each lot shall first have passed the individual tests. Acceptance of each lot will be determined by the launchers successfully passing the sampling tests, as specified in 4.3.2. All inspection and testing at the contractor's plant shall be under the supervision of the Government Inspector. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

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4.3.1 Individual Tests - Individual tests shall be conducted at the contractor's plant on each equipment submitted for acceptance under the contract, and shall consist of the following:

(1) Examination of Product (See 4.4.1)

(2) Operational Tests (See 4.4.2)

4.3.2 Sampling Tests - Sampling tests shall be conducted at a laboratory designated by the Bureau of Aeronautics on a quantity of equipments specified in 4.3.2.1. Equipments submitted for sampling tests shall be selected by the Government Inspector and first shall have passed the individual tests. Sampling tests shall include all the tests specified in 4.4 and any other tests deemed necessary by the Bureau of Aeronautics to determine that production equipment continues to be equivalent in performance and construction to the approved preproduction model.

4.3.2.1 Sampling Test Schedule - Sampling for acceptance testing shall be in accordance with the appendix to MIL-STD-105, inspection level L6, AQL as specified in 4.5.

4.3.2.2 Lot Sizes - The maximum quantity of launchers required to be delivered monthly as specified in the contract.

4.4 Tests

4.4.1 Examination of Product - Each equipment shall be subjected to a complete examination for the following to determine conformance with this specification.

(1) Dimensions

(2) Surface finishes of all parts

(3) Assembly Check

(4) Workmanship

4.4.1.1 Material and Process Check - In addition to the requirements of 4.4.1, the preproduction samples and acceptance lot samples shall be subjected to a complete material and process check to insure use of required and certified materials and compliance with required and approved manufacturing processes. Also, the samples shall be weighed to determine compliance with 3.2.4.

4.4.2 Operational Tests -

4.4.2.1 Ignition System - Continuity of the electrical circuits comprising the ignition system shall be checked. The grounding circuit continuity and resistance shall also be checked. The resistance measured between any Detent Pawl (drawing 55A27B9) and the suspension lug threaded inserts (drawing 55A27C58) shall not exceed 0.1 ohm.

4.4.2.2 Intervalometer - The intervalometer of each launcher shall be checked in the Ripple and the Single firing positions to determine proper functioning in accordance with the operational requirements as specified on drawing 55A27J1.

4.4.2.3 Detents - Random pull tests shall be conducted on individual detents. At least one detent shall be pull-tested on each launcher. The detent shall release with a pull of 1,600 to 2,400 pounds total load, but shall not release with a pull of less than 1,600 pounds total load or fail to release with a pull of 2,400 pounds total load.

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4.4.3 Waterproofness Test - The launcher in configuration "A" shall be tested in accordance with the waterproofness test of Specification MIL-P-116.

4.4.4 Moisture Penetration Resistance Test - Approximately one pound of completely reactivated desiccant in a bag shall be placed in the launcher. The launcher shall be conditioned for 24 hours at 100°F ±2°F and 95% relative humidity. After conditioning, the bag of desiccant shall be replaced with a fresh weighed bag of completely reactivated desiccant (approximately 1 pound). The launcher shall then be exposed for a period of 64 to 68 hours, to the conditions set forth above, after which it shall be reopened and the bag of desiccant reweighed. The weight increase shall not exceed .07 grams per 100 square inches of external launcher surface area (A) per 24 hours and shall be calculated as shown below:

$$\text{Transmission Rate} - \frac{(W2-W1)\ 2400}{T \times A}$$

W1 - Weight (in grams) at the beginning of the exposure period.
W2 - Weight (in grams) at the end of the exposure period.
T - Exposed period in hours (64 to 68 hours).
A - Area (total exposed in square inches.)

4.4.5 Environmental Testing -

4.4.5.1 High Temperature Test - The launcher in Configuration "A" (see 1.1) shall be placed within a heating chamber and the internal temperature of the chamber raised to +71°C with an internal relative humidity of not more than 5% and retained at this condition for a continuous period of 50 hours. At the end of the conditioning period, an ignition system continuity and resistance check shall be made, with the intervalometer removed, in accordance with Drawing 55A27J3. The grounding circuit continuity and resistance shall also be checked. The resistance measured between any Detent Pawl, (Drawing 55A27B9) and the suspension lug threaded inserts (Drawing 55A27C58) shall not exceed 0.1 ohm. The temperature shall then be reduced to room temperature (approximately 70°F (21°C)). A waterproofness test shall be conducted in accordance with 4.4.3.

4.4.5.2 Low Temperature Test - The launcher in Configuration "A" (see 1.1) shall be placed in a cooling chamber and the internal temperature of the chamber reduced to and maintained at -53.9°C for a continuous period of 24 hours. At the end of this conditioning period, ignition system and ground circuit continuity and resistance tests shall be conducted in accordance with 4.4.2.1. The temperature shall be raised to room temperature (approximately 20°C) and a waterproofness test shall then be conducted in accordance with 4.4.3.

4.4.5.3 Salt Spray Test - Salt spray tests shall be performed in accordance with the salt spray test procedure of Specification MIL-E-5272. The launcher in Configuration "A", (see 1.1) shall be subjected to salt spray for a period of 50 hours. At the end of the salt spray period, ignition system and grounding circuit continuity and resistance tests shall be conducted in accordance with 4.4.2.1. The interior of the launcher shall be inspected for moisture penetration. If evidence of moisture penetration is found, a waterproofness test shall be conducted in accordance with 4.4.3, followed by a pressure retention test in accordance with 4.4.4.

4.4.6 Vibration Test

4.4.6.1 Flight Vibration Test - The launcher is Configuration "B" (see 1.1) loaded with four inert 5.0-inch Folding Fin Aircraft Rockets, complete with inert warheads and fuzes, shall be subjected to translational vibration in each of the three principal axes in turn. The launcher shall be vibrated for a minimum period of one-half hour through a frequency range of 8 to 300 cps during which a search shall be made for conditions of resonance or complex motion. If a condition of resonance or complex motion is found, the launcher shall be vibrated for a period of 15 minutes at each frequency in which the condition occurred. If either resonance or complex motion occur at more than one frequency in any one of the three principal axes, the launcher shall be vibrated at the frequency which causes the most severe loading. The total excursion of vibration shall be held constant at 0.03 inch or shall result in a vibratory acceleration of 3.8 G whichever value is limiting. The electrical contact between the launcher and the rocket motor shall be monitored continuously during the three flight vibration tests. The complete firing circuit shall be checked for functionability, intermittently during each of the three flight

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vibration tests. Failure of the electrical ignition system to function as required shall be cause for rejection. The detents shall not release, break or in any other way allow the rockets to free themselves as a result of the flight vibration tests. At the conclusion of the tests the rockets and the interior of the launcher shall be inspected for excessive chafing or other damage resulting from the tests. The launcher shall remain intact with no visual exterior or interior damage.

4.4.6.2 Shipping Vibration - The loaded launcher in Configuration "A" of 1.1 shall be subjected to vibration test in accordance with ASTM-D-999-48T Procedure A (See 6.3). The time of vibration in each direction shall be for a period of not less than 30 minutes. The launcher shall suffer no damage that will prevent it from performing its intended functions. This test shall be performed on the same launcher which previously has been subjected to the test of 4.4.7.

4.4.7 Rough Handling Test - The loaded launcher in Configuration "A" (see 1.1) shall be subjected to a rough handling corner drop tests as follows: The launcher shall be supported at one corner on its base on a block 5 inches in height. A 12-inch block shall be placed under the adjacent corner of the same end. The opposite end of the launcher shall then be raised to a height of 24 inches and allowed to fall freely onto a solid steel or concrete surface. This test shall be performed once on each end. The launcher shall suffer no damage that will prevent it from performing its intended functions.

4.4.8 Static Loading Tests - In accordance with Specification MIL-A-8591 the launcher shall be subjected to static load tests as follows:

4.4.8.1 Center Section - The launcher center section, in the unloaded condition, shall be suspended from a bomb rack, or simulated bomb rack, having a 14-inch or 30-inch suspension. The test loads shall be applied in such a manner that the loads are equally divided among the four launching tubes and uniformly distributed over the length of the tubes. Three load forces shall be applied simultaneously at a point as near as possible to the C. G. of the total mass. The loads applied and direction of force shall be as follows:

(1) 11.5 G downward.

(2) 7.5 G to the side

in each direction (left to right and right to left).

(3) 9.0 G forward

(4) 9.0 G rearward

4.4.8.1.1 The test shall be in two parts as follows:

(1) Down load and side load, right to left, combined with forward load.

(2) Down load and side load left to right combined with rearward load.

The launcher shall not deflect to the extent of taking a permanent set at any point or show any visible signs of damage as a result of this test.

4.4.8.2 Fairings - The fairings shall be subjected to static loads simulating aerodynamic loads as follows:

(1) Axial (externally applied) - 1,390 pounds total load, applied in such a manner that the peak load force is at the tip of the cone and gradually diminishes to zero at the base, with equal forces applied over the periphery of the cone.

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(2) Side (externally applied) - a side load of 437 pounds total load equally distributed from the tip of the cone to the base shall be applied in a manner that also distributes the load over approximately one-half of the circumference.

The fairings shall withstand the above loads without showing any visible signs of failure or damage other than a slight pull of the material at the fairing to fairing band rivets.

4.4.9 Detent Tests -

4.4.9.1 Pull Test - Pull tests shall be conducted on individual detents. The detent shall release with a pull of 1,600 to 2,400 pounds total load, but shall not release with a pull of less than 1,600 pounds total load or fail to release with a pull of 2,400 pounds total load.

4.4.9.2 Shock Load Test - The launcher center section, loaded with four inert 5.0-inch Folding Fin Aircraft Rockets, complete with inert warheads and fuzes, shall be subjected to the forward and aft shock loads experienced during catapult take-off and arrested landing. The load applied shall be 13.5 G in each direction. The time duration of the shock load shall be a minimum of 30 milliseconds to a maximum of one second. The detents shall not break or release rockets during or as a result of this test.

4.4.10 Ground Firing Test - The loaded launcher complete with warheads and fuzes (warheads and fuzes may be inert) in Configuration "B" (see 1.1) shall be suspended from a bomb rack or simulated bomb rack having 14-inch or 30-inch suspension provisions and test fired. The ignition connection may be made through either the forward or aft receptacle. The ignition potential shall be between 18 and 29 volts DC. There shall be a 5-ohm limiting resistor in series with the launcher intervalometer. The launcher shall be capable of igniting all functional rockets in a normal manner and shall remain intact on the suspension device and sustain no damage other than peeling and darkening of the exposed surfaces.

4.4.11 Flight Test - The loaded launcher in Configuration "B" (see 1.1) shall be tested at a government facility, designated by the procuring service, under the following conditions:

(1) Dynamic pressure of 1,050 pounds per square foot at zero degrees yaw and plus 4 G.

(2) Dynamic pressure of 360 pounds per square foot at 8° right yaw and 8° left yaw.

4.5 Classification of Defects - Defects shall be classified and lots accepted or rejected on the basis of the acceptable Quality Level (AQL) for the particular classification as follows:

4.5.1 Critical Defects: AQL - 1.0%

(1) Failure of the detent mechanism to meet the requirements of 4.4.9.

(2) Failure of the launcher center section to meet the requirements of 4.4.8.

4.5.2 Major Defects: AQL - 1.0%

(1) Failure of the intervalometer to function in accordance with the operational requirements specified on Drawing 55A27D90.

(2) Failure of the electrical system to meet the continuity and resistance requirements of Drawing 55A27J3.

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(3) Failure of the launcher to pass any of the tests specified herein not listed under critical defects. (see 4.4 and 4.5.1).

4.5.3 Minor Defects: AQL - 4.0%

(1) Misalignment of forward and aft retainer rings.

(2) Improper fit and assembly of the shock pan, including the latch.

(3) Inadequate locking of the handle clips in either the extended or retracted position.

(4) Minor dents or burrs on the sealing edge of the retainer ring.

(5) Missing, loose, incorrect or illegible decalcomanias, placards or nameplates.

(6) Missing or improperly installed lockwires.

(7) Defective plating.

(8) Unsatisfactory painting (chalking, sagging, smearing, etc.)

(9) Inserts and/or lug threads not greased.

(10) Substandard workmanship, such as excessive weld splatters, injurious burrs, scratches, dents, etc.

(11) Any other defects, flaws, blemishes or missing minor parts not specifically listed herein, which in the opinion of the cognizant government inspector are a violation of Section 3.

NOTE: Although a defect may be classified as minor, at the time of inspection, it may be directly or indirectly responsible for the luancher failing to pass one of the tests specified under 4.4, in which case the defect classification shall be major or possibly critical. (i.e., minor dents or burrs on the sealing edge of the retainer ring may result in failure to pass the waterproofness test; improper fit and assembly of the shock pan may result in failure to pass the rough handling test, waterproofness test or both.)

4.6 Correction of Defects - Lots failing to pass inspection and testing shall be subjected to the requirements of MIL-STD-105, except as noted herein. Re-submitted samples failing to pass inspection and testing shall be cause for rejection of the lot represented by the samples. Such lots shall not be subjected to additional tests and shall be considered as unacceptable to the government. (If in the opinion of the contractor, the defects of the rejected lot are remediable, the contractor shall so indicate and outline in detail, in writing, to the procuring service.) Additional inspection and testing shall not be permitted unless specifically authorized by the procuring service in writing.

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5. PREPARATION FOR DELIVERY

5.1 Palletizing - The complete rocket launcher shall be palletized for delivery in accordance with procuring service instructions on government furnished pallets. Delivery shall be as directed by the procuring service.

5.2 Marking and Labeling -

5.2.1 Pallets - Marking of the pallets containing the rocket launcher packages shall be in accordance with requirements of MIL-STD-129.

6. NOTES

6.1 Availability of Dummy Stores for Testing - The contractor shall submit requests to the Bureau of Aeronautics for dummy and inert stores to be used for testing as required by this specification.

6.2 Adherence to Drawings and Specifications - Conflict Elimination - Insofar as the contractor is required by provisions of this specification to adhere rigidly to the drawings listed in Section 2, which may make it impossible in some instances also to comply with the performance and test requirements specified herein; then, in that event, the contractor should promptly notify the procuring activity of such fact, and appropriate changes will be made in the applicable drawings, parts, and/or performance requirements as authorized by the procuring activity. Resulting changes to the drawings or specifications shall be executed in accordance with 3.2.1.

6.3 American Society for Testing Materials Standard ASTM-D-999 Vibration Test for Shipping Containers.

(Copies of American Society for Testing Materials Standard may be obtained from the Society, 1916 Race Street, Philadelphia 3, Pennsylvania.)

NOTICE: When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility for any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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